

Войдем под пользователем user1:

```
root@eltex-practice2-v33:~$ su - user1
```

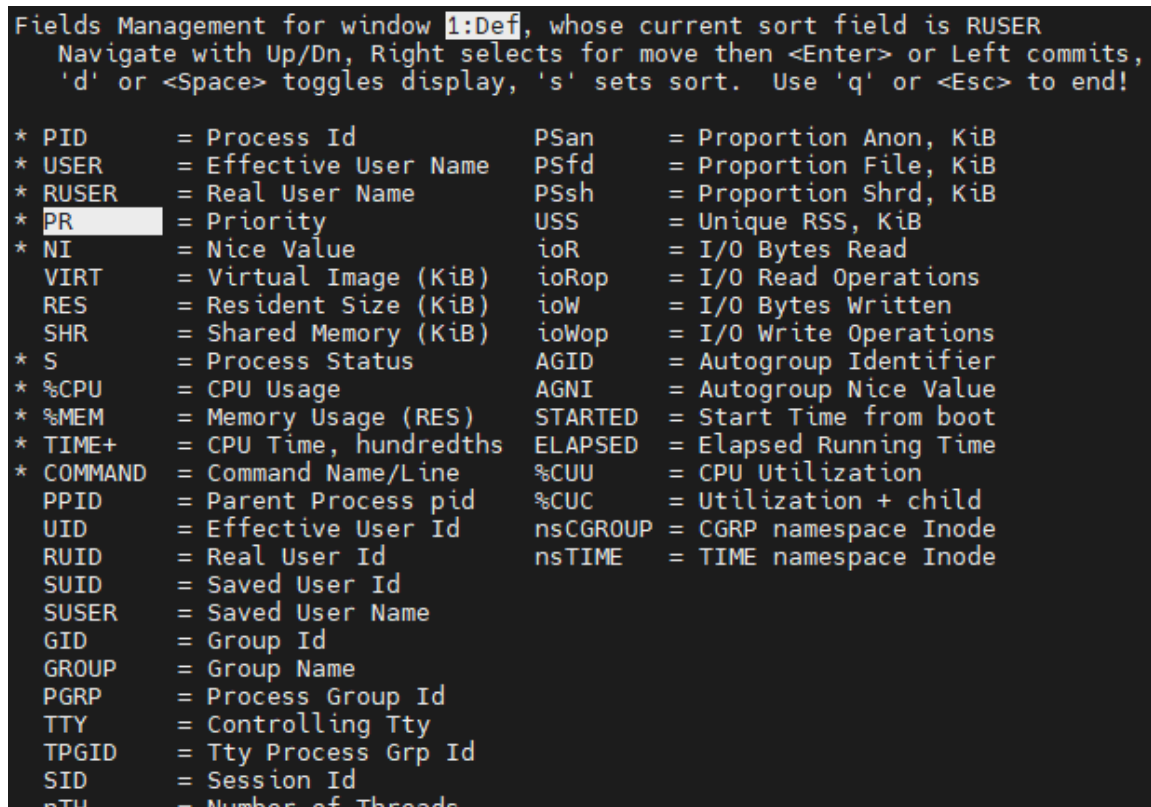
Подсчитаем количество процессов с более чем одним потоком:

```
user1@eltex-practice2-v33:~$ ps -eLf | awk 'NR > 1 && $6 > 1  
{print}' | wc -l
```

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Запустим top и настроим поля согласно заданию:

```
user1@eltex-practice2-v33:~$ top
```



```
Fields Management for window 1:Def, whose current sort field is RUSER
  Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
  'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!

* PID      = Process Id          PSan       = Proportion Anon, KiB
* USER     = Effective User Name PSfd       = Proportion File, KiB
* RUSER    = Real User Name     PSsh       = Proportion Shrd, KiB
* PR       = Priority           USS        = Unique RSS, KiB
* NI       = Nice Value         ioR        = I/O Bytes Read
  VIRT     = Virtual Image (KiB) ioRop      = I/O Read Operations
  RES      = Resident Size (KiB) ioW        = I/O Bytes Written
  SHR      = Shared Memory (KiB) ioWop      = I/O Write Operations
* S        = Process Status     AGID       = Autogroup Identifier
* %CPU     = CPU Usage          AGNI       = Autogroup Nice Value
* %MEM     = Memory Usage (RES) STARTED    = Start Time from boot
* TIME+    = CPU Time, hundredths ELAPSED    = Elapsed Running Time
* COMMAND  = Command Name/Line %CUU       = CPU Utilization
  PPID     = Parent Process pid %CUC       = Utilization + child
  UID      = Effective User Id  nsCGROUP   = CGRP namespace Inode
  RUID     = Real User Id       nsTIME     = TIME namespace Inode
  SUID     = Saved User Id
  SUSER    = Saved User Name
  GID      = Group Id
  GROUP    = Group Name
  PGRP     = Process Group Id
  TTY      = Controlling Tty
  TPGID    = Tty Process Grp Id
  SID      = Session Id
  nTH      = Number of Threads
```

Рисунок 1 – Настройка top

```

top - 04:32:54 up 13:23,  2 users,  load average: 0.00, 0.00, 0.00
Tasks: 115 total,   1 running, 114 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.2 sy,   0.0 ni, 99.7 id,   0.0 wa,   0.0 hi,   0.0 si,   0.0 st
MiB Mem :  3915.9 total,  3110.8 free,   417.4 used,   616.6 buff/cache
MiB Swap:  3185.0 total,  3185.0 free,    0.0 used.  3498.5 avail Mem

```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
42628	user1	user1	20	0	S	0.0	0.1	0:00.03	bash
49103	user1	user1	20	0	R	0.0	0.2	0:00.01	top
560	systemd+	systemd+	20	0	S	0.0	0.2	0:00.13	systemd-timesyn
559	systemd+	systemd+	20	0	S	0.0	0.3	0:00.09	systemd-resolve
543	systemd+	systemd+	20	0	S	0.0	0.2	0:00.35	systemd-network
711	syslog	syslog	20	0	S	0.0	0.2	0:00.05	rsyslogd
1	root	root	20	0	S	0.0	0.3	0:01.60	systemd
2	root	root	20	0	S	0.0	0.0	0:00.00	kthreadd
3	root	root	20	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	root	0	-20	I	0.0	0.0	0:00.06	kworker/0:0H-kblockd
12	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
14	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
15	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
16	root	root	20	0	S	0.0	0.0	0:00.07	ksoftirqd/0
17	root	root	20	0	I	0.0	0.0	0:00.54	rcu_preempt
18	root	root	rt	0	S	0.0	0.0	0:00.19	migration/0
19	root	root	-51	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	root	20	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	root	20	0	S	0.0	0.0	0:00.00	cpuhp/1
22	root	root	-51	0	S	0.0	0.0	0:00.00	idle_inject/1
23	root	root	rt	0	S	0.0	0.0	0:00.51	migration/1
24	root	root	20	0	S	0.0	0.0	0:00.12	ksoftirqd/1
26	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/1:0H-events_highpri
27	root	root	20	0	S	0.0	0.0	0:00.00	kdevtmpfs
28	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-inet_
29	root	root	20	0	S	0.0	0.0	0:00.00	kauditd
30	root	root	20	0	S	0.0	0.0	0:00.00	khungtaskd
33	root	root	20	0	S	0.0	0.0	0:00.00	oom_reaper
34	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-write
35	root	root	20	0	S	0.0	0.0	0:01.09	kcompactd0
36	root	root	25	5	S	0.0	0.0	0:00.00	ksmd
38	root	root	39	19	S	0.0	0.0	0:00.00	khugepaged
39	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-kinte
40	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-kbloc
41	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-blkcg
42	root	root	-51	0	S	0.0	0.0	0:00.00	irq/9-acpi
43	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-tpm_d
44	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-ata_s
45	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-md
46	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-md_bi
47	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-edac-
48	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-devfr
49	root	root	-51	0	S	0.0	0.0	0:00.00	watchdogd
50	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-quota
51	root	root	0	-20	I	0.0	0.0	0:00.07	kworker/1:1H-kblockd
52	root	root	20	0	S	0.0	0.0	0:00.00	kswapd0
53	root	root	20	0	S	0.0	0.0	0:00.00	ecryptfs-kthread
54	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-kthro
55	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-acpi_
57	root	root	20	0	S	0.0	0.0	0:00.00	scsi_eh_0
58	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-scsi_
59	root	root	20	0	S	0.0	0.0	0:00.00	scsi_eh_1
60	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-scsi_
61	root	root	20	0	S	0.0	0.0	0:00.00	scsi_eh_2

Рисунок 2 – Результат настройки top

Запустим в другом терминальном окне passwd:

```
root@eltex-practice2-v33:~$ su - user1
```

```
user1@eltex-practice2-v33:~$ passwd
```

```
root@eltex-practice2-v33:~# su - user1
user1@eltex-practice2-v33:~$ passwd
Changing password for user1.
Current password: █
```

Рисунок 3 – passwd

Перейдем в окно с top и отфильтруем процессы по реальному пользователю:

```
top - 04:36:52 up 13:26, 4 users, load average: 0.00, 0.00, 0.00
Tasks: 120 total, 1 running, 119 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 3104.7 free, 423.2 used, 616.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used, 3492.7 avail Mem
```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
42628	user1	user1	20	0	S	0.0	0.1	0:00.03	bash
49103	user1	user1	20	0	R	0.0	0.2	0:00.43	top
50269	user1	user1	20	0	S	0.0	0.1	0:00.01	bash
50291	root	user1	20	0	S	0.0	0.1	0:00.00	passwd

Рисунок 4 – Отфильтрованные процессы

Отправим процессу passwd следующие сигналы:

SIGTERM (15) – проигнорирован;

SIGINT (2) – проигнорирован;

SIGQUIT (3) – проигнорирован;

SIGKILL (9) – процесс убит;

```
user1@eltex-practice2-v33:~$ passwd
Changing password for user1.
Current password: Killed
user1@eltex-practice2-v33:~$ █
```

Рисунок 5 – Процесс принудительно остановлен

Откроем редактор и приостановим его работу сочетанием ctrl+z:

```
user1@eltex-practice2-v33:~$ vim ~/file_task3.txt
```

```
[1]+  Stopped
```

```
vim ~/file_task3.txt
```

```
top - 04:44:53 up 13:34, 4 users, load average: 0.00, 0.01, 0.00
Tasks: 123 total, 1 running, 121 sleeping, 1 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.2 sy, 0.0 ni, 99.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3915.9 total, 3089.7 free, 429.7 used, 625.5 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3486.2 avail Mem
```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
42628	user1	user1	20	0	S	0.0	0.1	0:00.03	bash
49103	user1	user1	20	0	R	0.3	0.2	0:00.75	top
50269	user1	user1	20	0	S	0.0	0.1	0:00.01	bash
53700	user1	user1	20	0	T	0.0	0.3	0:00.01	vim

Рисунок 6 – После приостановки vim

Аналогично поступим с командой sleep 600:

```
user1@eltex-practice2-v33:~$ sleep 600
```

```
^Z
```

```
[2]+  Stopped                  sleep 600
```

Проверим текущие задачи:

```
user1@eltex-practice2-v33:~$ jobs
```

```
[1]-  Stopped                  vim ~/file_task3.txt
```

```
[2]+  Stopped                  sleep 600
```

Возобновим sleep в фоне:

```
user1@eltex-practice2-v33:~$ bg
```

```
[2]+  sleep 600 &
```

Изменим у процесса sleep параметр nice на 10:

```
user1@eltex-practice2-v33:~$ renice -n 10 -p $(pgrep -f "sleep 600")
```

```
54156 (process ID) old priority 0, new priority 10
```

PID	USER	RUSER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
54156	user1	user1	30	10	S	0.0	0.1	0:00.00	sleep

Рисунок 7 – После изменения параметра nice

Сделаем vim активным и закроем его:

```
user1@eltex-practice2-v33:~$ fg  
vim ~/file_task3.txt
```

Отправим процессу sleep SIGTERM (15) и проверим задачи:

```
user1@eltex-practice2-v33:~$ kill -15 54156  
[2]+  Terminated                  sleep 600  
user1@eltex-practice2-v33:~$ jobs
```

Для текущего интерпретатора команд создадим перехватчик SIGINT (2) и SIGQUIT (3) :

```
user1@eltex-practice2-v33:~$ trap 'echo "Меня голыми руками не  
возьмешь!"' SIGINT SIGQUIT
```

Протестируем перехватчик:

```
user1@eltex-practice2-v33:~$ kill -3 $$  
Меня голыми руками не возьмишь!  
user1@eltex-practice2-v33:~$ kill -2 $$  
Меня голыми руками не возьмишь!
```